

MohammadHadiAlizadeh | Curriculum Vitae

I +989146901097 • # m.hadializa@gmail.com • § MohammadHadiAlizadeh

Education

Amirkabir University of Technology (Tehran Polytechnic)

M.Sc. in Chemical Engineering (Process Design) GPA:

18.63/20

Thesis: Model Predictive Control of an Intensified Semicontinuous Distillation Column.

Amirkabir University of Technology (Tehran Polytechnic)

B.Sc. in Chemical Engineering

GPA: 18.12/20

Ranked second among students.

Tehran, Iran

2021–2023

Tehran, Iran

2017–2021

Experience

Process Systems Engineering Lab, Tehran's Polytechnic

Process Engineer

Tehran, Iran

2024–Present

Simulated the air separation unit (ASU) of a petrochemical complex, including adsorption units, using Adsim.

Developed a dynamic ASU model with surge controllers for the MAC and BAC compressors and estimated missing plate-fin cold-box parameters (Linde design) through MATLAB/HYSYS Dynamic automation. Applied the validated information to perform dynamic start-up and shut-down scenario simulations for the unit.

Developed software for dynamic RO simulation.

National Elites Foundation

Simulation Engineer

Iran

2023–2024

Modeled a dynamic fuel injection line and machinery in DAE and compared the simulation results with practical results.

Amirkabir University of Technology (Tehran Polytechnic)

Research Assistant

Worked on multiple optimal control algorithms including Model Predictive Control.

Amirkabir University of Technology (Tehran Polytechnic)

Teaching Assistant

Chemical Process Safety (Teaching HAZOP) and Chemical Process Control Lab.

Tehran, Iran

2021–2023

Tehran, Iran

2021–2023

Industrial Projects

Simulation and Control: Air Separation Unit (ASU)

Developed the core dynamic model for an Operator Training Simulator (OTS), including steady-state and dynamic simulation of unit operations; implementation of key control loops (e.g., surge controller) and ESD logic; and sequential dynamic optimization to estimate missing parameters (e.g., Cold-box UA) using operating data.

Simulation and Feasibility Study: Pentane & Hexane Separation from C5+ Feed Plant

Conducted a feasibility study on pentane–hexane separation to supply solvents for HDPE and LLDPE reactors. Evaluated multiple base technologies through mass and energy balances and preliminary equipment rating, and compared economic performance using NPV and IRR to identify the optimal technology and feed option.

Feasibility Study: UV-based Ballast Water Treatment System

Investigated the feasibility of a UV-based ballast water treatment system by evaluating available technologies, performing a preliminary design through reverse engineering, and estimating the overall system cost.

Volunteer Contribution

OpenModelica Software Development Team

Member

2023–2024

Debugged and tested multiple OpenModelica features (e.g., OMMatlab) to enhance advanced process control capabilities in simulation environments. Contributed to reducing linearization time, achieving a 420× speed-up; critical for fast control applications.

GitHub: MohammadHadiAlizadeh

Chemical Engineering Research and Design

Peer Reviewer

Jan 2025

Reviewed and evaluated multiple technical papers for a leading journal.

Skills

Languages: English (Advanced)

Technical: Programming: Modelica (OOP/API), Julia(Numerical Calculations), Python (Machine learning libraries, Mathematical libraries), Matlab

Modeling & Control: First Principle Models, Data-driven Models (Linear Subspace, Neural Networks), Economic and Nonlinear Model Predictive Control (Moving Horizon, Shrinking Horizon)

Process Design: Chemical Process Design (Vessels, Distillation towers), Optimization, Aspen HYSYS(Dynamic, Steady), Aspen AdSim (PSA, TSA, Linde TPSA), Aspen Plus, OpenModelica, Dymola

Achievements & Awards

Ranked second among B.Sc. students The Best M.Sc Thesis of University 2024.

Research & Publications

2025: Mohammad Hadi Alizadeh, Ali M. Sahlodin, **Monoclonal Antibodies Manufacturing in a Fed-batch Bioreactor using Economic Nonlinear Model Predictive Control with Extended Kalman Filter**, *IJCCE*, 44 (01), 298-308, Iran.

2024: Mohammad Hadi Alizadeh, Ali M. Sahlodin, **Energy-efficient semi-continuous distillation of a ternary mixture using combined tracking-economic model predictive control**, *Chemical Engineering Research and Design Journal*, Volume 209, Pages 283-298, United Kingdom.

2023: Mohammad Hadi Alizadeh, Ali M. Sahlodin, Arunkumar Palanisamy, Francesco Casella, Peter Fritzon, **Application of the OpenModelica-Matlab Interface to Integrated Simulation and Successive Linearization Based Model Predictive Control**, presented at the 15th International Modelica Conference, Germany.

2023: *Mohammad Hadi Alizadeh, Ali M. Sahlodin*, **Shrinking-horizon Economic Model Predictive Control of a Fed-batch Reactor for Optimal MAb Production**, presented at the 12th International Chemical Engineering Congress & Exhibition, Iran.

Certifications

Data Science: Data Science, 365 Careers.

Reinforcement Learning: Reinforcement Learning, Loop Academy.

Equipment Design and Data-sheet: Falat Ghareh.

PFD/P&ID: AIPCECO.

Aspen Plus: Aspen Plus Course Certification, WR Training Pro.

Aspen HYSYS: Aspen Hysys Course Certification, Chemeng Home.

Economic Theories: Introduction to Economic Theories, Erasmus University Rotterdam.

Politics and economics of international energy: Sciences Po.